

1. Demand Side

Suppose there are N consumers, each endowed with a type γ_r for $r \in \{H, L\}$, corresponding to high or low utility from variety in products consumed. As such, there are $N_r \in N$ consumers of each type, $r \in \{H, L\}$. Consumers maximize utility according to the following:

$$U_{irjt} = \beta_0 + \gamma_r \mathbb{1}\{j_t \neq j_{t-1}\} + \alpha p_{rjt} + \varepsilon_{irjt}$$

Where utility is over individuals $i \in N$, type $r \in \{H, L\}$, products $j \in J$, and time $t \in T$. β_0 is an intercept, representing the baseline utility from consuming. γ_i is the consumer's utility from consuming different products, and is active if they switched products from one period to the next. α is disutility of price, which can be lowered via a discount from advertising. Finally, ε_{ijt} is the Gumbel error, giving us a closed form Logit.

This equation can be written in terms of deterministic vs. random utility:

$$\begin{aligned} U_{irjt} &= V_{irjt} + \varepsilon_{irjt} \\ \text{s.t.} \\ V_{irjt} &= \beta_0 + \gamma_r \mathbb{1}\{j_t \neq j_{t-1}\} + \alpha(p_{jt} - d_{rjt}) \end{aligned}$$

Next, choice probabilities of consuming the inside option(s) is written as such:

$$P_{irjt} = \frac{e^{V_{irjt}}}{1 + \sum_k e^{V_{irkj}}}$$

Where we normalize the outside option utility to be zero, with a choice probability of:

$$P_{ir0t} = \frac{1}{1 + \sum_k e^{V_{irkj}}}$$

Next, because consumers of the same type r are homogenous, we can say:

$$s_{rjt} = P_{irjt} \text{ for } r \in \{H, L\}$$

Taking the derivative with respect to prices, we can get the own and cross price elasticities:

$$\begin{aligned}\epsilon_{jj}^p &= \alpha p_{jt}(1 - s_{rjt}) \\ \epsilon_{jk}^p &= -\alpha p_{kt} s_{rkt} \quad \text{for } k \neq j\end{aligned}$$

Thus, since $\alpha < 0$, we can see that own price elasticity is negative and cross-price is positive.

2. Supply Side

Now, a supermarket is in the market. The supermarket is able to control prices $\bar{p}_j$ for a product j . The supermarket can also offer a discount d_{rjt} which is tailored to our switching types $r \in \{H, L\}$. The firm's actual price is written as:

$$\begin{aligned}p_{rjt} &\equiv \bar{p}_j - d_{rjt} \\ &\text{s.t.} \\ d_{rjt} &> 0\end{aligned}$$

So, as we raise the discount, we raise utility as $\alpha < 0$ so $-\alpha d_{rjt} > 0$. It is through this discounting that firm can induce switching. The firm's profit margin for a product j is given by

$$\begin{aligned}m_{rjt} &= p_{rjt} - c_j \\ &= \bar{p}_j - d_{rjt} - c_j\end{aligned}$$

This says that as we raise the discount, we raise market share on the product being discounted but lose profit margin.

With these definitions in mind, we can get into two scenarios. We will be looking at a two period model in, first, the case where the firm has perfect information on the consumers' types and can thus perfectly price. In the second case, we will look at the case of imperfect information, where the firm is using Bayesian updating to approximate the type distribution.

2.1. Perfect Information

Here, we assume the firm has perfect information over the distribution of types. That is, the supermarket perfectly observes each consumer type r and thus knows the mass N_r of each

type. Define period one profit, π_1 as such:

$$\begin{aligned}\pi_1 &= \sum_r N_r \sum_j s_{rj1}(m_{rj1}) \\ &\text{s.t.} \\ s_{rj1} &= P_{irj1} \\ m_{rj1} &= \bar{p}_j - d_{rj1} - c_j\end{aligned}$$

Here, the firm is understanding how each type's market share for each product impacts their market share. We can similarly define period two profits, π_2 like so:

$$\pi_2 = \sum_r N_r \sum_j s_{rj1} \left[\sum_k s_{rk2}(j)(m_{rk2}(j)) \right]$$

Here, the only addition is the firm now understands all of the possible margins for a consumer of type r who chose product j in period 1, and product k in period 2. For the sake of conciseness in notation, $w_r(j) = \sum_k s_{rk2}(j)(m_{rk2}(j))$. We can now write total profit, Π ,

$$\begin{aligned}\Pi &= \pi_1 + \pi_2 \\ &= \sum_r N_r \sum_j s_{rj1}(m_{rj1} + \delta w_r(j))\end{aligned}$$

Where δ is a discount factor. Taking the derivative with respect to period one prices for a product j yields:

$$\frac{\partial \Pi}{\partial p_{rj1}} : 0 = 1 + \alpha \sum_{k \neq j} s_{rk1} [(m_{rj1} - m_{rk1}) + \delta(w_r(j) - w_r(k))]$$

Here, the firm sets prices according to the current margin difference and the future profit difference between switching a consumer between product j vs. product k . When we do the same for period two, we recover the following:

$$\frac{\partial \Pi}{\partial p_{rk2}} : 0 = 1 + \alpha \sum_{l \neq k} s_{rl2}(j) [(m_{rk2}(j) - m_{rl2}(j))]$$

Here, the firm again thinks of the tradeoff, without having to think about future profit

as period two is terminal. Under perfect information, the supermarket chooses its prices by evaluating the current margin and future profit associated with reallocating consumers across products. Now, we move to the case of imperfect information.

2.2. Imperfect Information

In this case, the supermarket does not know consumer types with certainty, nor do they know the mass of either type. So, they must infer the distribution from consumer purchases. Define the following objects:

$$\begin{aligned}\lambda_1 &= Pr(r = H) \\ \lambda_2 &= Pr(r = H|j) \Rightarrow \lambda_2(j) = \frac{\lambda_1 s_{Hj1}}{\lambda_1 s_{Hj1} + (1 - \lambda_1) s_{Lj1}} \\ \tilde{s}_{j1} &= \lambda_1 s_{Hj1} + (1 - \lambda_1) s_{Lj1} \\ \tilde{s}_{k2}(j) &= \lambda_2(j) s_{Hk2}(j) + (1 - \lambda_2(j)) s_{Lk2}(j)\end{aligned}$$

Now, we can define the profit functions much like we did before:

$$\begin{aligned}\pi_1 &= N \sum_j \tilde{s}_{j1} m_{j1} \\ \pi_2 &= N \sum_j \tilde{s}_{j1} \sum_k \tilde{s}_{k2}(j) m_{k2}(j)\end{aligned}$$

The key difference here is that the firm must now update their beliefs from period to period rather than being able to discount according to the tradeoffs between two products. This implies that there is room for error in discounting. When combining the two into overall profit, we get:

$$\Pi = N \sum_j \tilde{s}_{j1} [m_{j1} + \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j)]$$

Now, we can take the derivative with respect to period one prices for product j :

$$\begin{aligned}
\frac{\partial \Pi}{\partial p_{j1}} : 0 = & N\tilde{s}_{j1} + \alpha N[\lambda_1 s_{Hj1}(1 - s_{Hj1}) + (1 - \lambda_1)s_{Lj1}(1 - s_{Lj1})][m_{j1} + \delta \sum_k \tilde{s}_{k2}(j)m_{k2}(j)] \\
& - \alpha N \sum_{h \neq j} [\lambda_1 s_{Hh1}s_{Hj1} + (1 - \lambda_1)s_{Lh1}s_{Lj1}][m_{h1} + \delta \sum_k \tilde{s}_{k2}(h)m_{k2}(h)] \\
& + \delta \sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} [\pi_{H2}(h) - \pi_{L2}(h)]
\end{aligned}$$

This is much the same as the perfect information case. However, looking at the third line, we can see where the firm incorporates their belief updating. Define the derivative in the third line as such:

$$\frac{\partial \lambda_2(h)}{\partial p_{j1}} = \begin{cases} 0 = \frac{\alpha \lambda_1 (1 - \lambda_1)}{\tilde{s}_{j1}^2} [s_{Hj1}s_{Lj1}(s_{Lj1} - s_{Hj1})] & \text{if } h = j \\ 0 = \frac{\alpha \lambda_1 (1 - \lambda_1)}{\tilde{s}_{h1}^2} [s_{Hh1}s_{Lh1}(s_{Lj1} - s_{Hj1})] & \text{if } h \neq j \end{cases}$$

So, the firm updates its belief about the distribution of consumer types upon observing period one consumption. The magnitude and direction of this update depends on how likely that choice is for each type. In total, the third line of the derivative interacts the probability of purchasing item h with the effect of price on the firm's posterior belief and the value of believing in one type or the other.

3. Solving for Optimal Prices

I can solve for optimal prices in both cases, beginning with the simpler perfect information specification.

3.1. Perfect Information

We begin from our derivative with respect to prices as recovered in the previous section. Remember that is given by:

$$0 = 1 + \alpha \left[\sum_{k \neq j} s_{rk1} (m_{rj1} - m_{rk1}) + \delta (w_r(j) - w_r(k)) \right]$$

s.t.

$$m_{rj1} = p_{rj1} - c_j$$

$$m_{rk1} = p_{rk1} - c_k$$

Now, expanding the expression and solving:

$$\begin{aligned} 0 &= 1 + \alpha \left[\sum_{k \neq j} s_{rk1} ((p_{rj1} - c_j) - (p_{rk1} - c_k) + \delta (w_r(j) - w_r(k))) \right] \\ 0 &= 1 + \alpha (p_{rj1} - c_j) \sum_{k \neq j} s_{rk1} - \alpha \sum_{k \neq j} s_{rk1} (p_{rk1} - c_k) + \alpha \delta \sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k)) \\ \alpha (p_{rj1} - c_j) \sum_{k \neq j} s_{rk1} &= -1 + \alpha \sum_{k \neq j} s_{rk1} (p_{rk1} - c_k) - \alpha \delta \sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k)) \\ \alpha p_{rj1} \sum_{k \neq j} s_{rk1} &= \alpha c_j \sum_{k \neq j} s_{rk1} - 1 + \alpha \sum_{k \neq j} s_{rk1} (p_{rk1} - c_k) - \alpha \delta \sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k)) \\ p_{rj1}^* &= c_j - \frac{1}{\alpha \sum_{k \neq j} s_{rk1}} + \frac{\sum_{k \neq j} s_{rk1} (p_{rk1} - c_k)}{\sum_{k \neq j} s_{rk1}} - \delta \frac{\sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k))}{\sum_{k \neq j} s_{rk1}} \\ p_{rj1}^* &= c_j - \frac{1}{\alpha (1 - s_{rj1})} + \frac{\sum_{k \neq j} s_{rk1} m_{rk1}}{1 - s_{rj1}} - \delta \frac{\sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k))}{1 - s_{rj1}} \end{aligned}$$

From this, the firm with perfect information sets each type r 's price to equal the cost plus some logit markup plus the share weighted average margin of alternatives $k \neq j$ minus the intertemporal tradeoff. Here, the firm will be willing to lower the price of good j today if it implies higher profits tomorrow ($w_r(j) > w_r(k)$), increasing probability of purchase. In the opposite case, the firm would raise prices on product j to encourage shifting from product j to product k tomorrow. That is, they use the past choice to determine consumer state and thus future profits for each product.

3.2. Imperfect Information

$$\begin{aligned}
0 = & N\tilde{s}_{j1} + \alpha N[\lambda_1 s_{Hj1}(1 - s_{Hj1}) + (1 - \lambda_1)s_{Lj1}(1 - s_{Lj1})][m_{j1} + \delta \sum_k \tilde{s}_{k2}(j)m_{k2}(j)] \\
& - \alpha N \sum_{h \neq j} [\lambda_1 s_{Hh1}s_{Hj1} + (1 - \lambda_1)s_{Lh1}s_{Lj1}][m_{h1} + \delta \sum_k \tilde{s}_{k2}(h)m_{k2}(h)] \\
& + \delta \sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} [\pi_{H2}(h) - \pi_{L2}(h)]
\end{aligned}$$

s.t.

$$\begin{aligned}
m_{j1} &= p_{j1} - c_j \\
\lambda_1 &= \Pr(r = H) \\
\lambda_2 &= \Pr(r = H|j) \Rightarrow \lambda_2(j) = \frac{\lambda_1 s_{Hj1}}{\lambda_1 s_{Hj1} + (1 - \lambda_1)s_{Lj1}} \\
\tilde{s}_{j1} &= \lambda_1 s_{Hj1} + (1 - \lambda_1)s_{Lj1} \\
\tilde{s}_{k2}(j) &= \lambda_2(j)s_{Hk2}(j) + (1 - \lambda_2(j))s_{Lk2}(j) \\
\pi_2 &= N \sum_j \tilde{s}_{j1} \sum_k \tilde{s}_{k2}(j)m_{k2}(j) \\
\mathbb{A} &= [\lambda_1 s_{Hj1}(1 - s_{Hj1}) + (1 - \lambda_1)s_{Lj1}(1 - s_{Lj1})]
\end{aligned}$$

Now, we can solve for p_{j1} via the following steps. First,

$$\begin{aligned}
\alpha N \mathbb{A} [p_{j1} - c_j + \delta \sum_k \tilde{s}_{k2}(j)m_{k2}(j)] = \\
- N\tilde{s}_{j1} + \alpha N \sum_{h \neq j} [\lambda_1 s_{Hh1}s_{Hj1} + (1 - \lambda_1)s_{Lh1}s_{Lj1}][m_{h1} + \delta \sum_k \tilde{s}_{k2}(h)m_{k2}(h)] \\
- N\delta \sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} [\pi_{H2}(h) - \pi_{L2}(h)]
\end{aligned}$$

Now, we isolate p_{j1} :

$$\begin{aligned}
\alpha N \mathbb{A} p_{j1} &= \alpha N \mathbb{A} c_j - \alpha N \mathbb{A} \delta \sum_k \tilde{s}_{k2}(j)m_{k2}(j) \\
& - N\tilde{s}_{j1} + \alpha N \sum_{h \neq j} [\lambda_1 s_{Hh1}s_{Hj1} + (1 - \lambda_1)s_{Lh1}s_{Lj1}][m_{h1} + \delta \sum_k \tilde{s}_{k2}(h)m_{k2}(h)] \\
& - N\delta \sum_h \tilde{s}_{h1} \left[\frac{\partial \lambda_2(h)}{\partial p_{j1}} (\pi_{2H}(h) - \pi_{2L}(h)) \right]
\end{aligned}$$

And simplify:

$$\begin{aligned}
p_{j1} = & c_j - \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j) - \frac{\tilde{s}_{j1}}{\alpha A} \\
& + \frac{\sum_{h \neq j} [\lambda_1 s_{Hh1} s_{Hj1} + (1 - \lambda_1) s_{Lh1} s_{Lj1}]}{A} [m_{h1} + \delta \sum_k \tilde{s}_{k2}(h) m_{k2}(h)] \\
& - \frac{\delta}{\alpha A} \left[\sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} (\pi_{2H}(h) - \pi_{2L}(h)) \right]
\end{aligned}$$

So, breaking down the optimal price term-by-term, we have that optimal price is made up of the marginal cost, minus the own-product discounted continuation value, plus a markup term and the value of alternative products, and lastly a term referring to the learning adjustment from period one behavior. In the imperfect-information specification, today's prices affect current demand and the distribution of future states as well as the information that can be learned from consumer purchases. Period one prices alter the likelihood of purchase, thus affecting the posterior over consumer type and, by extension, optimal pricing in period two.

4. Difference from Two Period Pricing

I will briefly show how this differs from a normal two-period pricing game without state dependence. The perfect information case is easier to show this with, so I will proceed with it. Recall from the optimal pricing:

$$p_{rj1}^* = c_j - \frac{1}{\alpha(1 - s_{rj1})} + \frac{\sum_{k \neq j} s_{rk1} m_{rk1}}{1 - s_{rj1}} - \delta \frac{\sum_{k \neq j} s_{rk1} (w_r(j) - w_r(k))}{1 - s_{rj1}}$$

In the traditional pricing game, the firm would not track past purchases, only types r . Thus, period two demand does not depend on period one purchases. That is, $w_r(j) = w_r(k)$ and we would lose the last term resulting in

$$p_{rj1}^* = c_j - \frac{1}{\alpha(1 - s_{rj1})} + \frac{\sum_{k \neq j} s_{rk1} m_{rk1}}{1 - s_{rj1}}$$

Or marginal cost plus a markup plus the value of alternatives. To put it simply, the profit function for the traditional game is:

$$\Pi = \pi_1(p_{rj1}) + \delta\pi_2(p_{rj2})$$

s.t.

$$\frac{\partial \pi_2}{\partial p_{rj1}} = 0$$

Where prices can be set independently of each other. In the model proposed above, period two is directly influenced by period one since the firm is keeping track of past purchases.